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Parallel Governance
Decentralized Autonomous
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***Abstract*—Decentralized autonomous organizations(DAOs) enabled by blockchain and smart contracts is regarded as an effective tool to solve corporate governance problems. It can minimize the contract risks, principal-agent dilemmas, and other issues in corporate governance. However, DAOs also face a variety of governance issues, such as internal governance issues and external governance issues caused by the technical attributes of blockchain and smart contracts. DAO is a typical complex system with the characteristics of both social and engineering complexity. It is difficult for existing governance theories to solve governance problems in DAOs. The parallel intelligence theory based on the artificial systems + computational experiments + parallel execution (ACP) method is a suitable research and practical approach to solve management and control problems for complex systems. Therefore, we propose a parallel governance framework for DAO based on the parallel intelligence theory and discuss its key research issues. Finally, we construct a parallel governance system for GnosisDAO, and conduct computational experiments on its voting mechanism. The results show the defects of GnosisDAO governance mechanism and we thus propose further research directions. We consider that parallel governance represents the future trend of DAO research and practice. This paper is aimed at stimulating further effort and providing helpful guidance and reference for future research works.**

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***Index Terms*—I**
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